Spectral Representations and Computational Modeling for Time-Series Prediction

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Abstract

This paper studies time-series prediction using Fourier analysis, STL decomposition, and machine learning methods.

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1 Introduction

2 Fourier Ratio

2.1 Functions on $\mathbb{Z}_n$

The set of integers is denoted by

$$\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}.$$

$\mathbb{Z}_n$ is also a set of numbers, but $\mathbb{Z}_n$ represents a collection of sets consisting of integers that have the same remainder.

When some integers have the same remainder after being divided by the same number n, these integers are considered equivalent and can be classified into the same set. This equivalence can be written as

$$x \equiv y \pmod{n}.$$

According to the rules of division, when the divisor is n, there are n possible remainders:

$$0, 1, 2, \ldots, n-1.$$

For example, when $n = 3$, $\mathbb{Z}_3$ contains three equivalence classes, [0], [1], and [2], which correspond to

$$[0] = \{\ldots, -6, -3, 0, 3, 6, 9, \ldots\},$$

$$[1] = \{\ldots, -5, -2, 1, 4, 7, \ldots\},$$

$$[2] = \{\ldots, -4, -1, 2, 5, 8, \ldots\}.$$

Within each of the sets [0], [1], and [2], all integers are equivalent to one another.

In general, $\mathbb{Z}_n$ can be understood as the set formed by dividing all integers into n equivalence classes according to their remainders when divided by n.

In $\mathbb{Z}_n$, after the remainder reaches $n-1$, adding 1 causes it to return to 0:

$$0 \to 1 \to 2 \to \cdots \to n-1 \to 0 \to \cdots$$

This forms a cycle.

For example, in $\mathbb{Z}_3$,

$$1 + 2 = 3,$$

and

$$3 \equiv 0 \pmod{3}.$$

Therefore, in $\mathbb{Z}_3$,

$$1 + 2 \equiv 0 \pmod 3.$$

This demonstrates the cyclic structure of $\mathbb{Z}_n$. We can think of $\mathbb{Z}_n$ as moving around a circle.

Now we need a function that transforms this cyclic structure into the form of a unit circle. This function is called a character, defined by

$$\chi(x) = e^{\frac{2\pi i x}{n}}.$$

Since $\cos^2\theta + \sin^2\theta = 1$, by Euler's formula,

$$e^{i\theta} = \cos\theta + i\sin\theta,$$

we can view $e^{i\theta}$ as a point $(\cos\theta, \sin\theta)$ on the complex plane. Since this point satisfies $x^2 + y^2 = 1$, it lies on the unit circle.

We know that one complete circle has an angle of 2π. To place the n positions of $\mathbb{Z}_n$ evenly on the unit circle, we divide the entire circle into n equal parts. Therefore, $\frac{2\pi}{n}$ is the angle of each step around the circle.

When we reach the x-th position, the total angle is

$$\frac{2\pi x}{n}.$$

Therefore,

$$e^{\frac{2\pi i x}{n}}$$

represents the point on the unit circle corresponding to the x-th position in $\mathbb{Z}_n$.

2.2 Discrete Fourier Transform

A linear combination is formed by multiplying each element in a collection by a coefficient and then adding the results together. For example, if $\mathbf{u}$ and $\mathbf{v}$ are vectors, then

$$\mathbf{w} = a\mathbf{u} + b\mathbf{v}$$

is a linear combination of $\mathbf{u}$ and $\mathbf{v}$, where a and b are coefficients.

The same idea can be applied to functions. For each $m \in \mathbb{Z}_n$, define the function

$$\chi_m(x) = e^{\frac{2\pi i m x}{n}}.$$

The functions χ_m form the Fourier basis for complex-valued functions on $\mathbb{Z}_n$. Therefore, a function

$$f : \mathbb{Z}_n \to \mathbb{C}$$

can be expressed as a linear combination of these Fourier basis functions:

$$f(x) = \sum_{m \in \mathbb{Z}_n} c_m \chi_m(x) = \sum_{m \in \mathbb{Z}_n} c_m e^{\frac{2\pi i m x}{n}},$$

where c_m is the coefficient of the basis function χ_m.
The coefficient c_m is calculated by

$$c_m = \frac{1}{n} \sum_{x \in \mathbb{Z}_n} f(x) e^{-\frac{2\pi i m x}{n}}.$$

This coefficient is also written as $\widehat{f}(m)$. Therefore, the discrete Fourier transform of f is defined by

$$\widehat{f}(m) = \frac{1}{n} \sum_{x \in \mathbb{Z}_n} f(x) e^{-\frac{2\pi i m x}{n}}.$$

The value $\widehat{f}(m)$ measures the contribution of the m-th Fourier basis function to the original function f.

2.3 L^1, L^2, and L^p Norms

A norm is a way to measure the size of a vector or a function. For a function defined on $\mathbb{Z}_n$,

$$f : \mathbb{Z}_n \to \mathbb{C},$$

since $\mathbb{Z}_n$ contains n elements, we can view the function f as a vector with n components:

$$f = (f(0), f(1), \ldots, f(n-1)).$$

We can measure the size of the function by calculating the norm of this vector.

2.3.1 L^1 Norm

First, we define the L^1 norm as

$$\|f\|_1 = \sum_{x \in \mathbb{Z}_n} |f(x)|.$$

We use absolute values so that the function values do not cancel each other out. This ensures that each function value contributes a nonnegative amount to the calculation.

For example, let

$$f(0) = 1, \quad f(1) = -2, \quad f(2) = 3.$$

Then,

$$\begin{aligned}\|f\|_1 &= |1| + |-2| + |3| \\ &= 1 + 2 + 3 \\ &= 6.\end{aligned}$$

Therefore, the L^1 norm measures the sum of the absolute values of all the function values.

2.3.2 L^2 Norm

Based on the L^1 norm, we can also square the absolute value of each function value, add them together, and finally take the square root. This gives us the L^2 norm.

We define the L^2 norm as

$$\|f\|_2 = \left(\sum_{x \in \mathbb{Z}_n} |f(x)|^2 \right)^{1/2}.$$

For example, using the previous function

$$f = (1, -2, 3),$$

we obtain

$$\begin{aligned}\|f\|_2 &= \left(|1|^2 + |-2|^2 + |3|^2\right)^{1/2} \\ &= \sqrt{1 + 4 + 9} \\ &= \sqrt{14}.\end{aligned}$$

In fact, the L^2 norm has the same form as the Euclidean norm we learned in linear algebra.

For example, for a two-dimensional vector

$$v = (a, b),$$

its length can be expressed as

$$\|v\|_2 = \sqrt{|a|^2 + |b|^2}.$$

Therefore, we can understand the L^2 norm as an application of Euclidean length to functions.

2.3.3 L^p Norm

By observing the definitions of the L^1 and L^2 norms, we can see that for the L^1 norm, we calculate the sum of the absolute values of the function values. For the L^2 norm, we calculate the sum of their squares and then take the square root.

We can generalize these two definitions to the L^p norm:

$$\|f\|_p = \left(\sum_{x \in \mathbb{Z}_n} |f(x)|^p \right)^{1/p}, \qquad 1 \leq p < \infty.$$

Here, p is a real number greater than or equal to 1.

When $p = 1$,

$$\|f\|_1 = \sum_{x \in \mathbb{Z}_n} |f(x)|.$$

When $p = 2$,

$$\|f\|_2 = \left(\sum_{x \in \mathbb{Z}_n} |f(x)|^2 \right)^{1/2}.$$

Therefore, the L^1 and L^2 norms are special cases of the L^p norm.

As p increases, larger function values contribute more heavily to the sum relative to smaller values.

As p approaches infinity, we obtain the L^∞ norm:

$$\|f\|_\infty = \max_{x \in \mathbb{Z}_n} |f(x)|.$$

The L^∞ norm represents the largest absolute value among all the function values.

2.3.4 Relationship Between L^p and L^∞ Norms

We already know that the definition of the L^p norm is

$$\|f\|_p = \left(\sum_{x \in \mathbb{Z}_n} |f(x)|^p \right)^{1/p},$$

and the definition of the L^∞ norm is

$$\|f\|_\infty = \max_{x \in \mathbb{Z}_n} |f(x)|.$$

Now, we want to prove that as p approaches infinity, the L^p norm approaches the L^∞ norm. We want to show that

$$\lim_{p \to \infty} \|f\|_p = \|f\|_\infty.$$

We can use the Squeeze Theorem to prove this result. We can find an upper bound and a lower bound for $\|f\|_p$, and then show that both bounds approach the same limit.

First, let

$$M = \max_{x \in \mathbb{Z}_n} |f(x)| = \|f\|_\infty.$$

Since M is the maximum absolute value of the function, for every $x \in \mathbb{Z}_n$, we have

$$|f(x)| \leq M.$$

Raising both sides to the power p, we get

$$|f(x)|^p \leq M^p.$$

Then, we sum over all $x \in \mathbb{Z}_n$:

$$\sum_{x \in \mathbb{Z}_n} |f(x)|^p \leq \sum_{x \in \mathbb{Z}_n} M^p.$$

Since $\mathbb{Z}_n$ contains n elements and M is a constant, the right-hand side can be written as nM^p. Therefore,

$$\sum_{x \in \mathbb{Z}_n} |f(x)|^p \leq nM^p.$$

Taking the $1/p$ power of both sides, we obtain

$$\left(\sum_{x \in \mathbb{Z}_n} |f(x)|^p \right)^{1/p} \leq (nM^p)^{1/p}.$$

By the definition of the L^p norm, we get

$$\|f\|_p \leq n^{1/p} M.$$

Thus, we have found an upper bound for $\|f\|_p$. We also need to find a lower bound.

Since M is the maximum absolute value of the function and $\mathbb{Z}_n$ is a finite set, there must exist some $x_0 \in \mathbb{Z}_n$ such that

$$|f(x_0)| = M.$$

We know that

$$\sum_{x \in \mathbb{Z}_n} |f(x)|^p$$

is the sum of $|f(x)|^p$ over all $x \in \mathbb{Z}_n$, and this sum must contain the term $|f(x_0)|^p = M^p$.

Since all the other terms are nonnegative, we have

$$\sum_{x \in \mathbb{Z}_n} |f(x)|^p \geq M^p.$$

Taking the $1/p$ power of both sides

$$\left(\sum_{x\in\mathbb{Z}_n} |f(x)|^p\right)^{1/p} \geq M.$$

Therefore,

$$\|f\|_p \geq M.$$

Now that we have found both the upper and lower bounds, we can combine them:

$$M \leq \|f\|_p \leq n^{1/p}M$$

Let p approaching infinity, n is a fixed positive integer, we know that

$$\lim_{p\to\infty} n^{1/p} = 1.$$

Therefore,

$$\lim_{p\to\infty} n^{1/p}M = M.$$

On the other hand, M on the left-hand side is a constant, so its limit is also M.

By the Squeeze Theorem, since both the upper and lower bounds approach M, the expression $\|f\|_p$ between them must also approach M, so that

$$\lim_{p\to\infty} \|f\|_p = M.$$

Since we defined

$$M = \|f\|_\infty,$$

we can conclude that

$$\lim_{p\to\infty} \|f\|_p = \|f\|_\infty$$

Therefore, we have proved that as p approaches infinity, the L^p norm approaches the L^∞ norm.

2.3.5 Two Special Cases of Hölder's Inequality

We want to further study the relationship between the norms of two functions when they are multiplied together.

Suppose that

$$f, g : \mathbb{Z}_n \to \mathbb{C}.$$

We want to prove the following two inequalities:

$$\|fg\|_1 \leq \|f\|_1 \|g\|_\infty$$

and

$$\|fg\|_1 \leq \|f\|_2 \|g\|_2.$$

These two inequalities are special cases of Hölder's Inequality.

Part 1: Proof of the First Inequality By the definition of the L^1 norm, we know

$$\|fg\|_1 = \sum_{x \in \mathbb{Z}_n} |f(x)g(x)|.$$

Since the absolute value of a product is equal to the product of the absolute values, we can write

$$\|fg\|_1 = \sum_{x \in \mathbb{Z}_n} |f(x)||g(x)|.$$

Next, we know

$$\|g\|_\infty = \max_{x \in \mathbb{Z}_n} |g(x)|.$$

Since $\|g\|_\infty$ is the maximum absolute value of the function g, for every $x \in \mathbb{Z}_n$, we have

$$|g(x)| \leq \|g\|_\infty.$$

Multiplying both sides by $|f(x)|$, and since the absolute value is nonnegative, the direction of the inequality does not change:

$$|f(x)||g(x)| \leq |f(x)|\|g\|_\infty.$$

Then, we sum over all $x \in \mathbb{Z}_n$:

$$\sum_{x \in \mathbb{Z}_n} |f(x)||g(x)| \leq \sum_{x \in \mathbb{Z}_n} |f(x)|\|g\|_\infty.$$

Since $\|g\|_\infty$ is a constant and does not depend on x, we can take it outside the summation:

$$\sum_{x \in \mathbb{Z}_n} |f(x)|\|g\|_\infty = \|g\|_\infty \sum_{x \in \mathbb{Z}_n} |f(x)|.$$

By the definition of the L^1 norm,

$$\sum_{x \in \mathbb{Z}_n} |f(x)| = \|f\|_1.$$

Therefore, we obtain

$$\|fg\|_1 \leq \|g\|_\infty \|f\|_1.$$

In other words,

$$\|fg\|_1 \leq \|f\|_1 \|g\|_\infty.$$

Thus, we have proved the first inequality.

Part 2: Proof of the Second Inequality Next, we want to prove that

$$\|fg\|_1 \leq \|f\|_2 \|g\|_2.$$

This time, we can use the Cauchy–Schwarz Inequality to prove the result.

According to the Cauchy–Schwarz Inequality, for two finite sequences a_i and b_i, we have

$$\sum_{i=1}^{n} |a_i||b_i| \leq \left(\sum_{i=1}^{n} |a_i|^2 \right)^{1/2} \left(\sum_{i=1}^{n} |b_i|^2 \right)^{1/2}.$$

First, we start with the definition of the L^1 norm:

$$\|fg\|_1 = \sum_{x \in \mathbb{Z}_n} |f(x)g(x)|.$$

Using the property of absolute values, we can write

$$\|fg\|_1 = \sum_{x \in \mathbb{Z}_n} |f(x)||g(x)|.$$

Now, let

$$a_x = |f(x)|, \qquad b_x = |g(x)|.$$

Then, we can directly apply the Cauchy–Schwarz Inequality to obtain

$$\sum_{x \in \mathbb{Z}_n} |f(x)||g(x)| \leq \left(\sum_{x \in \mathbb{Z}_n} |f(x)|^2 \right)^{1/2} \left(\sum_{x \in \mathbb{Z}_n} |g(x)|^2 \right)^{1/2}.$$

Next, we return to the definition of the L^2 norm.

We know that

$$\|f\|_2 = \left(\sum_{x \in \mathbb{Z}_n} |f(x)|^2 \right)^{1/2}$$

and

$$\|g\|_2 = \left(\sum_{x \in \mathbb{Z}_n} |g(x)|^2 \right)^{1/2}.$$

Therefore, we can rewrite the inequality as

$$\|fg\|_1 \leq \|f\|_2 \|g\|_2.$$

Thus, we have proved the second inequality.

Part 3: Relationship Between the Two Inequalities Now, we have proved that

$$\|fg\|_1 \leq \|f\|_1 \|g\|_\infty$$

and

$$\|fg\|_1 \leq \|f\|_2 \|g\|_2.$$

Although these two inequalities involve different norms, they both study how we can use the norms of two functions to estimate the L^1 norm of their product.

In fact, both inequalities are special cases of Hölder's Inequality.

Hölder's Inequality tells us that when $p, q \in [1, \infty]$ satisfy

$$\frac{1}{p} + \frac{1}{q} = 1,$$

where $1/\infty = 0$, we have

$$\|fg\|_1 \leq \|f\|_p \|g\|_q.$$

When $p = 1$ and $q = \infty$, we obtain

$$\|fg\|_1 \leq \|f\|_1 \|g\|_\infty.$$

When $p = q = 2$, we obtain

$$\|fg\|_1 \leq \|f\|_2 \|g\|_2.$$

Therefore, we can see that these two inequalities come from the same more general result.

Through these two proofs, we have not only studied the relationship between different L^p norms, but also learned how to use the norms of individual functions to estimate the norm of their product.

This also provides a foundation for our next step, where we will study norm inequalities involving convolution.

2.3.6 Minkowski's Inequality

2.3.7 Riesz–Thorin Interpolation Theorem

2.4 Fourier Ratio

2.4.1 Convolution Theorem

We want to compute

$$\widehat{f * g}(m).$$

This expression represents the Fourier transform of the convolution $f * g$. We know that the Fourier transform of a function h is defined as

$$\widehat{h}(m) = \frac{1}{N^d} \sum_x \chi(-m \cdot x) h(x).$$

Substituting $h(x) = (f * g)(x)$ into this definition, we obtain

$$\widehat{f * g}(m) = \frac{1}{N^d} \sum_x \chi(-m \cdot x)(f * g)(x).$$

Since $(f * g)(x)$ is a convolution, it has its own definition:

$$(f * g)(x) = \sum_y f(y) g(x - y).$$

Substituting this definition into the previous expression gives

$$\widehat{f * g}(m) = \frac{1}{N^d} \sum_x \chi(-m \cdot x) \sum_y f(y) g(x - y).$$

To simplify this expression, let

$$u = x - y.$$

Then,

$$g(x - y) = g(u) \qquad \text{and} \qquad x = y + u.$$

Therefore,

$$\begin{aligned} \chi(-m \cdot x) &= \chi\big(-m \cdot (y + u)\big) \\ &= \chi(-m \cdot y - m \cdot u). \end{aligned}$$

Since

$$\chi(a + b) = \chi(a)\chi(b),$$

we have

$$\chi\big(-m \cdot (y + u)\big) = \chi(-m \cdot y)\chi(-m \cdot u).$$

Thus,

$$\begin{aligned}
\widehat{f * g}(m) &= \frac{1}{N^d} \sum_x \chi(-m \cdot x) \sum_y f(y) g(x-y) \\
&= \frac{1}{N^d} \sum_{u,y} \chi\big(-m \cdot (y+u)\big) f(y) g(u) \\
&= \frac{1}{N^d} \sum_{u,y} \chi(-m \cdot y) \chi(-m \cdot u) f(y) g(u) \\
&= \frac{1}{N^d} \sum_{u,y} \Big[\chi(-m \cdot y) f(y)\Big] \Big[\chi(-m \cdot u) g(u)\Big] \\
&= \frac{1}{N^d} \left(\sum_y \chi(-m \cdot y) f(y)\right) \left(\sum_u \chi(-m \cdot u) g(u)\right).
\end{aligned}$$

The two factors have the same form as the sums in the definitions of the Fourier transforms, except for the normalization factor $\frac{1}{N^d}$. In particular,

$$\widehat{f}(m) = \frac{1}{N^d} \sum_y \chi(-m \cdot y) f(y)$$

and

$$\widehat{g}(m) = \frac{1}{N^d} \sum_u \chi(-m \cdot u) g(u).$$

Therefore, we introduce the necessary normalization factors:

$$\begin{aligned}
\widehat{f * g}(m) &= \frac{1}{N^d} \left(\sum_y \chi(-m \cdot y) f(y)\right) \left(\sum_u \chi(-m \cdot u) g(u)\right) \\
&= N^d \left(\frac{1}{N^d} \sum_y \chi(-m \cdot y) f(y)\right) \left(\frac{1}{N^d} \sum_u \chi(-m \cdot u) g(u)\right).
\end{aligned}$$

Therefore,

$$\boxed{\widehat{f * g}(m) = N^d \widehat{f}(m) \widehat{g}(m)}$$

The previous calculation showed that the Fourier transform turns convolution into multiplication. We now begin with ordinary multiplication and take its Fourier transform to determine whether it becomes a convolution.

We consider $f(x)g(x)$ as a new function $h(x)$:

$$h(x) = f(x)g(x).$$

We then take the Fourier transform of this function:

$$\widehat{fg}(m) = \frac{1}{N^d} \sum_x \chi(-m \cdot x) f(x) g(x).$$

Our goal is to rewrite this expression in terms of the Fourier transforms of f and g. To make $\widehat{f}(k)$ appear, we use the Fourier inversion formula:

$$f(x) = \sum_k \widehat{f}(k)\chi(k \cdot x).$$

Substituting this formula for $f(x)$ into the previous expression, we obtain

$$\widehat{fg}(m) = \frac{1}{N^d} \sum_x \chi(-m \cdot x) \left(\sum_k \widehat{f}(k)\chi(k \cdot x) \right) g(x).$$

We can move the sum over k outside and group the terms depending on x:

$$\widehat{fg}(m) = \sum_k \widehat{f}(k) \left(\frac{1}{N^d} \sum_x \chi(-m \cdot x)\chi(k \cdot x)g(x) \right).$$

Since

$$\chi(a)\chi(b) = \chi(a+b),$$

we have

$$\begin{aligned} \chi(-m \cdot x)\chi(k \cdot x) &= \chi(-m \cdot x + k \cdot x) \\ &= \chi\big(-(m-k) \cdot x\big). \end{aligned}$$

Therefore,

$$\widehat{fg}(m) = \sum_k \widehat{f}(k) \left(\frac{1}{N^d} \sum_x \chi\big(-(m-k) \cdot x\big) g(x) \right).$$

The expression inside the parentheses is the Fourier transform of g evaluated at $m-k$:

$$\widehat{g}(m-k) = \frac{1}{N^d} \sum_x \chi\big(-(m-k) \cdot x\big) g(x).$$

Thus,

$$\widehat{fg}(m) = \sum_k \widehat{f}(k)\widehat{g}(m-k).$$

By the definition of convolution,

$$(\widehat{f} * \widehat{g})(m) = \sum_k \widehat{f}(k)\widehat{g}(m-k).$$

Therefore,

$$\boxed{\widehat{fg}(m) = (\widehat{f} * \widehat{g})(m)}$$

This shows that the Fourier transform turns multiplication into convolution.

2.4.2 Convolution Inequality

3 STL

4 Machine Learning Model

5 Conclusion